

Software Testing Documentation

*Development of a Rice Supply Chain Management Application
for Tuan May Business*

StockLite - Report 5.0

Project Name	Development of a rice supply chain management application for Tuan May Business
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Change Log

Version	Date	Author	Summary
1.0	03/08/2026	Nguyễn Đình Đức	Official Test Plan migrated to the new university TestDoc template. Reviewer: Dương Nguyễn Danh Nhân.
1.1	03/08/2026	Nguyễn Đình Đức	Final reconciliation with SRS v2.1.0, RTW v2.1, FDS v2.3.0 and TDS v2.1.0. Corrected NFR wording, JOB-01..JOB-06, DataChangeHub scope, UAT coverage and baseline gaps. Reviewer: Dương Nguyễn Danh Nhân.
1.2	04/08/2026	Nguyễn Đình Đức	Reconciled Test Plan with runtime code: added JOB-07 Debt Due and Overdue Reminder coverage, corrected FT-level UAT/planned-coverage semantics, documented L3 business-flow coverage, and documented manual Auditor-role setup. Reviewer: Dương Nguyễn Danh Nhân.
1.3	07/08/2026	Nguyễn Đình Đức	Aligned the Test Type × Level Matrix with the documented strategy: Input Validation remains at L1/L2 and Background Job Testing remains within L2. Confirmed planned L1/L2/L3 coverage for FT-01..FT-06 and updated the active FDS baseline to v2.4.0 and SRS baseline reference to v2.2.0. Reviewer: Dương Nguyễn Danh Nhân.
1.4	12/08/2026	Nguyễn Đình Đức	Reconciled internal Test Plan inconsistencies: aligned performance gates with NFR-P01..P04; consolidated L3 System/API and business-flow coverage into Report 5.3 only; normalized workbook names; clarified L2 blocked-case closure rules, background-job criteria, execution summary metrics and defect terminology. Reviewer: Dương Nguyễn Danh Nhân.
1.5	18/08/2026	Nguyễn Đình Đức	Reconciled workbook cross-references with the actual delivered filenames: SU26_StockLite_Report_5.1_UnitTests_L1.xlsx, SU26_StockLite_Report_5.2_IntegrationTests_L2.xlsx, Report 5.3_SystemTests_L3_StockLite.xlsx and SU26_StockLite_Report_5.4_UAT.xlsx (previously misnamed in Sections 2.2 and 4.1). Added FT-26 Customer Feedback & Quality Traceability to the in-scope functional requirements (1.1) and Requirements Coverage Matrix (4.2) to match SRS v2.2.0/FDS v2.6.0. Updated the Section 1.5 baseline-control references to Vision & Scope v2.2.1 and FDS v2.6.0. Reviewer: Dương Nguyễn Danh Nhân.
1.6	18/08/2026	Nguyễn Đình Đức	Reference refresh (same version, no scope change): updated the Section 1.5 baseline-control row to cite Vision & Scope v2.2.2, SRS v2.2.1 and RTW v2.3, replacing the stale v2.2.1/v2.2.0/v2.1 citations left after those documents were renumbered/patched later the same day.

Version	Date	Author	Summary
1.7	22/08/2026	Nguyễn Đình Đức	Updated Report 5.2 Integration Test execution from the verified LOCAL_ONLY Wave 9 workbook: 905 total cases, 812 Pass, 53 Fail, 40 Blocked and 0 Not Run. Synchronized L2 execution, defect, security, environment, account-fixture and evidence descriptions. No password or token is stored in this document.
1.8	23/08/2026	Nguyễn Đình Đức	Synchronized Report 5.2 Integration Test results after final status closure: 905 total cases, 843 Pass, 62 Fail, 0 Blocked and 0 Not Run. Updated controller/API, Scheduler, Security, execution-summary and defect references.

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1. Scope of Testing

1.1 In Scope - Functional Requirements

Code	Feature / Requirement	Priority
FT-01	Paddy Procurement Scheduling	Must
FT-02	Paddy Purchase Receipt & Payable Debt	Must
FT-03	Farmer / Paddy-Seller Master	Must
FT-04	Lot-Column-Zone Warehousing	Must
FT-05	Quality Management & Preservation	Should
FT-06	Quarantine & Sale Lock	Should
FT-07	Milling Order (Paddy-to-Rice Conversion)	Must
FT-08	Dynamic Milling Yield Configuration	Should
FT-09	Lot-Based Product Costing	Must
FT-10	Multi-State Inventory	Must
FT-11	Lot Traceability	Must
FT-12	Column/Lot Outbound Strategy & Put-away	Should
FT-13	Stocktake by Zone/Column/Lot	Should
FT-14	Rule-Based Quality / Loss / Shrinkage Alerts	Should
FT-15	Audit Trail / Activity Log	Must
FT-16	Direct Call-In Sales Order (requires milling)	Must
FT-17	Scheduled Wholesale / Contract Sales	Should
FT-18	Customer & Receivable Debt	Must
FT-19	Packing/Weighing (BLE/manual) & Outbound Dispatch	Must
FT-20	Returns & Quality Handling (Reverse Logistics)	Should
FT-21	Role-Based Access Control	Must
FT-22	Multi-Warehouse & Internal Transfer	Should
FT-23	Rice Management Reports	Must
FT-24	Mobile Field Data-Entry App	Should
FT-25	BLE Scale Integration (ESP32/HX711)	Should
FT-26	Customer Feedback & Quality Traceability	Should

1.2 In Scope - Non-Functional Requirements

Code	Category	NFR Description & Target
NFR-P01	Performance	Dashboard summary and stock-by-lot view load within 3 seconds for 10,000 inventory records under normal network conditions.
NFR-P02	Performance	QR scan / weight-attach validation returns within 2 seconds for 95% of requests.
NFR-P03	Realtime	Inventory and alert broadcasts reach connected web clients within 5 seconds after a confirmed transaction.

Code	Category	NFR Description & Target
NFR-P04	Performance	Mobile field lookup (farmer, lot, order) loads within 3 seconds under normal network conditions.
NFR-S01	Security	All authenticated APIs require JWT and enforce server-side role authorization for the seven internal roles, including the read-only Auditor / Internal Reviewer.
NFR-S02	Security	Passwords are stored using BCrypt or an equivalent salted hash; plain passwords are never stored or logged.
NFR-S03	Security	Mobile access and refresh tokens are stored only in platform-secure storage and never displayed or written to logs.
NFR-A01	Auditability	Every stock-changing and financial (debt) action records actor, timestamp, source document, lot and reason where applicable.
NFR-R01	Reliability	A staff-confirmed BLE/manual weight submitted by the mobile app is idempotent by CaptureRequestId or operation key, so retries cannot duplicate evidence or stock/debt posting.
NFR-R02	Reliability	Recurring background jobs for alerts and notifications are idempotent; re-running with unchanged input creates no duplicate outcome. BLE alerts and notifications are not backend jobs.
NFR-U01	Usability	Field screens (purchase, put-away, milling completion, packing) require no more than 3 primary taps after a valid scan or scale reading.
NFR-U02	Usability	Interactive mobile controls provide a minimum touch target of 44 x 44 px for use in the field/warehouse.
NFR-C01	Compatibility	The web admin supports current Chrome/Edge/Safari; the mobile app supports common Android devices used by staff.
NFR-M01	Maintainability	The backend follows Clean Architecture; dynamic rules (yield, thresholds, put-away weights) are DB-configurable without redeployment.
NFR-B01	Backup	A database backup/export procedure is documented for the deployment and demonstration environment.
NFR-L01	Localization	Primary UI copy supports Vietnamese rice-trade terminology; internal IDs and API fields remain English.
NFR-D01	Data Integrity	Concurrent inventory updates are serialized through optimistic locking (RowVersion); conflicting writes fail safely and retry, never producing a lost update or negative available quantity (single small business, low write concurrency).
NFR-SC01	Scalability	OrganizationId is present (nullable) on business tables so the schema is ready for multi-tenant SaaS; the Global Query Filter is not enabled in the MVP (one business, Tuan May).

1.3 Out of Scope

Item	Reason
LI-01 / LI-02	No route optimisation/live maps and no full accounting, tax or statutory ledger; only operational delivery and two-way debt/reporting are in MVP.
LI-03 / LI-04	No enterprise ERP modules and no AI/ML forecasting or autonomous optimisation; rule-based alerts and scoring only.

Item	Reason
LI-05 / LI-06	No public App Store/Play Store release and no Shopee, marketplace or online storefront synchronisation; controlled Android APK and offline sales only.
LI-07 / LI-08 / LI-09	No automated mill control, conveyors/robots/smart racks or certified laboratory quality system. Stale Screen Design items for standalone BLE-device management and SCR-30..SCR-35 are excluded unless formally re-baselined.

1.4 Constraints

Category	Constraint
Architecture	Single-business layered/modular monolith; all inventory, lot, debt and ledger mutations share one MySQL transaction boundary.
Hosting	Render free-tier single instance can cold-start for up to about 50 seconds; warm-up is required before tests/demos.
Mobile / hardware	Flutter Android controlled APK; ESP32/HX711 communicates only with Flutter over BLE. Scale capacity/calibration and iOS support remain open OI-01.
External services	Aiven MySQL SSL, Firebase FCM, Cloudinary, Gmail SMTP, SignalR and Hangfire must be reachable or replaced by approved controlled test doubles.
Team / schedule	Five-member team and a 14-week project schedule; iterative L1/L2 work runs in Weeks 6-11, system testing in Week 12, UAT in Week 13 and closing in Week 14.

1.5 Assumptions

#	Assumption
1	Vision & Scope v2.2.2 and SRS v2.2.1 control business scope; RTW v2.3 controls traceability; TDS v2.1.0 and FDS v2.6.0 control architecture, active screens, APIs and jobs.
2	The API, Angular 17 Web Admin, Flutter Android APK and BLE scale builds are deployable to the documented environments.
3	Seven internal roles, Vietnamese test data, MySQL seed/migration data, FCM tokens and resettable demo accounts are available.
4	Application logs, Render/Hangfire dashboards, MySQL evidence, DataChangeHub events and GitHub Issues are accessible to QA.
5	OI-01..OI-06 remain explicit execution dependencies and must be resolved or formally accepted before final UAT sign-off.
6	The RTW Business Rules sheet defines BR-01..BR-53 only. BR-54 and BR-55 are referenced by SRS/TDS/RTW but their rule rows are missing and must be reconciled before traceability approval.
7	ScreenDesignSpec metadata states v2.1.0 while its change log reaches v2.2.0. SCR-30..SCR-35 are supplemental/stale until formally re-baselined because FDS v2.4.0 and TDS define SCR-01..SCR-29 as active.
8	Runtime/code reconciliation: JobRegistrar.cs registers DebtDueAndOverdueReminderJob and the migration CompleteJob04DebtDueAndOverdueReminders exists, while FDS v2.4.0 documents only JOB-01..JOB-06. This plan tracks the running job as JOB-07 until it is formally disabled/de-scoped or added to the approved baseline.

2. Test Strategy

2.1 Testing Types

a. Functional Testing

Field	Content
Objective	Verify FT-01..FT-25 and SC-01..SC-14 across the .NET API, Angular Web Admin, Flutter Android app and BLE-assisted field workflows.
Technique	Use equivalence partitions, boundary values, state-transition coverage, BR/NAC/error-code checks, seven-role permission partitions, screen-state tests, current-browser compatibility and Android/BLE interruption scenarios. Regression is prioritized around stock, debt, lot traceability and idempotency.
Completion Criteria	All executed P1 and Critical/High-priority release-gate functional cases pass; every approved FT/AC/NAC/BV/BR/NFR in scope maps to planned or executed evidence as applicable; no Critical/High case remains Failed or Blocked without an approved written waiver; final open defects < 5 and defect fix rate >= 95%.

b. UI Testing

Field	Content
Objective	Verify every active FDS v2.4.0 screen within SCR-01..SCR-29 renders the correct state, actions, role visibility, feedback and navigation on Angular Web and Flutter Android.
Technique	Angular component/page tests, Flutter widget tests, L2 API-driven screen-state checks, browser/device compatibility checks, permission-state partitions and navigation assertions against the FDS.
Completion Criteria	Every active SCR-ID has L2 evidence for render/state/navigation and System Test evidence for its critical cross-screen flow; no open Critical/High UI defect.

c. Input Validation Testing

Field	Content
Objective	Verify invalid and boundary input is rejected consistently by Angular/Flutter clients and ASP.NET Core APIs with the documented message and error code.
Technique	Unit Test validator/service tests plus Integration Test two-direction checks: submit through the real UI/API path and bypass the client with Postman using the same invalid payload. Apply equivalence partitioning and boundary value analysis to SRS fields and business rules.
Completion Criteria	Every Must-priority field and documented boundary has server-side evidence; critical user-entry fields also have client-visible validation evidence; no accepted invalid value changes stock or debt.

d. Background Job Testing

Field	Content
Objective	Verify all seven background jobs registered in the runtime process only eligible records, remain idempotent and produce the expected database, alert or notification outcome: JOB-01 Low Stock Detection; JOB-02 Intake Bottleneck Evaluation; JOB-03 Lot Quality Recheck & Long-Stored Alert; JOB-04 Notification Retry; JOB-05 User Session Cleanup; JOB-06 Verification Token Cleanup; JOB-07 Debt Due and Overdue Reminder. JOB-07 is included for execution safety while the code/FDS baseline difference is formally resolved.
Technique	Invoke job logic directly with controlled MySQL state and timestamps; assert database state, retry behavior, duplicate prevention, logs and FCM/alert outcomes. Do not depend on the cron trigger for deterministic L2 tests.
Completion Criteria	All executable JOB-01..JOB-07 eligible, ineligible, boundary and idempotency paths have L2 evidence; ineligible records remain unchanged; reruns create no duplicate alerts, debt reminders, notifications or state transitions. Environment-dependent failure paths that cannot be executed are recorded as Blocked with the fixture/environment limitation and closure action documented.

e. Performance Testing

Field	Content
Objective	Verify NFR-P01..P04, realtime delivery and BR-22 concurrency on the documented MVP demo environment.
Technique	Use k6 or JMeter with 10,000 seeded inventory records and about 50 VUs; measure QR/weight validation, mobile lookup APIs and SignalR visibility. Include Render cold/warm behavior and competing reservations.
Completion Criteria	Dashboard summary and stock-by-lot view load within 3 s for the 10,000-record test dataset; QR scan / weight-attach validation p95 <= 2 s; mobile field lookup <= 3 s under normal network conditions; realtime/SignalR visibility <= 5 s after a confirmed transaction; measured request error rate < 1%; no lost update or negative available quantity.

f. Security Testing

Field	Content
Objective	Verify selected L2 authentication, server-side authorization and security controls against BR-45, BR-46, BR-47, BR-51, BR-55 and NFR-S01–S03. The scope is limited to controls with actual API, xUnit, source-review or log evidence.
Technique	Use automated Postman/HTTP requests against the local ASP.NET Core API for missing or malformed JWTs, wrong-role access, HTTP 401/403 behavior, logout/device ownership, selected audit and QR endpoints, and direct server-side validation bypass. Use xUnit service tests, server-log inspection and Swagger/backend-source comparison as supporting evidence. Controlled fixtures and evidence are recorded for every executed case.
Completion Criteria	All 19 Security_CrossCutting cases have execution status and evidence: 15 Pass, 4 Fail, 0 Blocked and 0 Not Run. Confirmed authorization defects are recorded by BUG-ID; completion does not imply zero Critical/High defects.

2.2 Test Levels

a. Test Type × Level Matrix

Type of Tests	L1 Unit	L2 Integration	L3 System	L4 UAT
Functional Testing	✓	✓	✓	✓
UI Testing	✓	✓	✓	✓
Input Validation Testing	✓	✓	-	-
Background Job Testing	-	✓	-	-
Performance Testing	-	-	✓	-
Security Testing	-	✓	✓	-

b. Level 1 - Unit Test

Field	Content
In-charge	Developers by component; QA owner: Nguyễn Đình Đức
When	During development and whenever relevant backend code is changed; affected L1 tests are re-executed before feature/release completion.
Focus	.NET backend application services, validators, helpers, extensions, business/state guards, inventory, costing and debt logic. External dependencies are mocked to maintain L1 test isolation. Tests requiring BackendContext or EF Core test databases are classified as L2 and excluded from the L1 report.
Structure	Tests are organized by SUT/component. Scenarios follow Given/When/Then and cover valid, invalid, boundary, equivalence-partition and decision/branch cases, with traceability primarily to UC/BR references. Each [Fact] is counted as one execution case, while each [Theory] is expanded by its [InlineData] cases.
Files	SU26_StockLite_Report_5.1_UnitTests_L1.xlsx
Tools	xUnit, Moq, FluentAssertions and MockQueryable.Moq for .NET backend unit testing.
Acceptance Criteria	All Critical/High-priority and state/business-rule unit tests pass; all recorded L1 execution cases complete without open Critical/High L1 defects. Unit-test execution results must show no regression in previously passing L1 cases.

c. Level 2 - Integration Test

Field	Content
In-charge	Nguyễn Đình Đức, with support from backend developers when necessary.
When	During development iterations and before the final system testing phase.
Focus	API request/response contracts, authentication and RBAC, CRUD, filtering/paging, server-side validation, physical-bag workflows, inventory reconciliation, state transitions, idempotency, background jobs and security cross-cutting controls. Updated API regression cases are included by affected module.

Field	Content
Structure	Controller/module cases use Given / When / Then and record role, validation direction, coverage technique, expected result, status, BUG-ID and evidence. Automation first validates route, body, role and fixture data; a backend defect is recorded only after collection and test-data errors are excluded. Multi-step workflows use API-created disposable fixtures and API cleanup.
Files	SU26_StockLite_Report_5.2_IntegrationTests_L2.xlsm
Tools	Postman collections, automated Node.js/PowerShell HTTP harnesses, Swagger/OpenAPI and backend-source comparison, xUnit integration/service tests, Hangfire trigger checks, API follow-up reads and controlled server-log inspection. Execution target: local backend http://127.0.0.1:5257 only; production/remote Swagger was not mutated or security-tested.
Acceptance Criteria	Current result: 905 total L2 cases; 843 Pass, 62 Fail, 0 Blocked and 0 Not Run (93.1% overall pass rate). Controller/API scope: 849 cases; 795 Pass, 54 Fail, 0 Blocked and 0 Not Run (93.6% pass rate). Scheduler: 37 cases; 33 Pass and 4 Fail. Security: 19 cases; 15 Pass and 4 Fail. Fail results require BUG-ID evidence.

Background Job Testing (Dimension 3) - within L2

Field	Content
In-charge	Backend developers with QA verification by Nguyễn Đình Đức
When	When background-job functionality is ready for integration testing.
Focus	JOB-01 Low Stock Detection; JOB-02 Intake Bottleneck Evaluation; JOB-03 Lot Quality Recheck & Long-Stored Alert; JOB-04 Notification Retry; JOB-05 User Session Cleanup; JOB-06 Verification Token Cleanup; JOB-07 Debt Due and Overdue Reminder.
Structure	Dedicated Scheduler/Background Jobs sheet: arrange database/time state -> invoke job logic -> assert database, logs and notifications -> rerun for idempotency.
Files	Included in SU26_StockLite_Report_5.2_IntegrationTests_L2.xlsm on dedicated job sheets.
Tools	xUnit integration tests, the available project database/test fixture, Hangfire job services and controlled FCM/log evidence.
Acceptance Criteria	All executable JOB-01..JOB-07 eligible, ineligible, boundary and idempotency paths pass; ineligible records remain unchanged; reruns create no duplicate alerts, debt reminders, notifications or state transitions. Environment-dependent failure paths that cannot be executed must be recorded as Blocked with the fixture/environment limitation and closure action documented before release sign-off.

d. Level 3 - System Test

3a. System / API Flow Testing

Field	Content
In-charge	Nguyễn Đình Đức; reviewed by Dương Nguyễn Danh Nhân

Field	Content
When	After backend deployment and integration completion; rerun affected system/API flows after relevant API, business-rule or database changes and before release completion.
Focus	End-to-end HTTP/API flows across major business processes; request/response contracts; authentication and RBAC; business/state-transition rules; validation and error handling; concurrency-sensitive inventory operations; realtime updates; security checks; and applicable system-level NFRs.
Structure	System/API test cases are organized by business flow. Each case defines the HTTP/API action, expected result, priority, execution status, defect reference and notes/evidence. Cross-domain flows cover the interaction of multiple APIs and business processes rather than isolated service methods.
Files	Report 5.3_SystemTests_L3_StockLite.xlsx
Tools	Postman (Manual) for HTTP/API execution; Swagger/OpenAPI for API contract reference; Browser DevTools or application logs where supporting evidence is required
Acceptance Criteria	All executed Critical/High-priority system/API release-gate cases pass; no Critical/High case remains Failed or Blocked without an approved waiver; applicable NFR/security gates meet their documented baselines; inventory, debt and state transitions remain consistent; final open defects < 5 and defect fix rate >= 95%.

3b. Web / Mobile Business Flow Testing (within L3 System Test)

Field	Content
In-charge	Nguyễn Đình Đức; review/support by Dương Nguyễn Danh Nhân
When	After the required Web, Mobile and backend functions for a business flow are available; rerun affected business-flow and regression-smoke cases after relevant functional changes and before release completion.
Focus	Cross-interface business journeys across Angular Web, Flutter Mobile and backend services, including authentication, paddy purchase, put-away, quality handling, milling, sales/outbound, stocktake, internal transfer, customer return, debt, reports, alerts, RBAC/Auditor behavior, QR/traceability and applicable cross-platform interactions.
Structure	Business-flow cases are organized within the L3 System Test workbook and executed through real user-facing Web/Mobile interfaces without business-layer mocks. Each flow defines actor/role, user action, expected business result, priority, execution status, defect reference and execution notes/evidence. Regression-smoke cases provide a shorter critical-path system check.
Files	Report 5.3_SystemTests_L3_StockLite.xlsx (same L3 workbook; no separate E2E workbook)
Tools	Manual testing on the deployed Web application and Android application/device or emulator; QR scanning and BLE-related behavior are verified on the applicable mobile flow; Postman may be used only as supporting API evidence when required.

Field	Content
Acceptance Criteria	All Critical/High-priority Web/Mobile business-flow and regression-smoke cases required for release pass; no critical workflow remains blocked without an approved waiver; inventory, lot, status and debt effects remain consistent across related processes; no duplicate posting or unauthorized write operation occurs; no open Critical/High defect blocks release.

e. Level 4 - UAT (User Acceptance Testing)

Field	Content
In-charge	Tuan May / mentor representative, supported by Nguyễn Đình Đức and Dương Nguyễn Danh Nhân
When	Week 13 after L1-L3 quality gates pass; final acceptance before Week 14 closing.
Focus	Real rice procurement, storage, milling, sales, debt, dispatch, returns, traceability and field usability outcomes.
Structure	Business-language scripts organized by SC-01..SC-14 with expected outcome, evidence and Pass/Conditional/Fail decision.
Files	SU26_StockLite_Report_5.4_UAT.xlsx
Tools	Manual walkthrough on Render staging, Angular Web, controlled Android APK and physical BLE scale.
Acceptance Criteria	All P1 UAT scripts pass; no critical business blocker; conditional items have owners/deadlines; written business sign-off obtained.

2.3 Supporting Tools

Purpose	Tool	Vendor	Version
API exploratory/manual testing	Postman	Postman Inc.	Current project version
API integration testing	Postman collections + HTTP automation harness	Postman Inc. / Project	Version recorded in execution evidence
Web component testing	Jasmine + Karma	Angular	Angular 17 baseline
Mobile widget testing	flutter_test	Flutter	Flutter 3 baseline
Backend unit testing	xUnit + Moq	.NET	.NET 8
Integration environment	Local ASP.NET Core API, current development database and EF Core InMemory service-integration fixtures	Microsoft / .NET / Project	.NET 8 / EF Core 8; database build recorded in evidence
Performance testing	k6 or JMeter	Grafana / Apache	Version to be recorded at execution
Security testing	Postman collections, PowerShell HTTP harness and xUnit authorization tests	Postman Inc. / Microsoft / .NET	Versions recorded in execution evidence

Purpose	Tool	Vendor	Version
Web system testing	Manual scripted tests on Angular Web Admin	Angular	Angular 17
Mobile system testing	Manual scripted tests on Flutter Android APK	Flutter	Flutter 3 / Dart 3
BLE hardware testing	ESP32 + HX711 BLE GATT; physical or simulated readings	Project hardware	Firmware/model to be confirmed OI-01
CI/CD pipeline	GitHub Actions (.github/workflows/dotnet.yml)	GitHub	Project baseline
Deployment	Docker multi-stage + Render.com	Docker / Render	Free-tier single instance
Database	Aiven managed MySQL with SSL required	Aiven	MySQL 8.0
Notifications / evidence	Firebase FCM, Cloudinary, SMTP, Gmail	Google / Cloudinary / Gmail	Provider API versions
Defect tracking	GitHub Issues	GitHub	Project baseline
Test management/reporting	Report 5 Excel workbooks + RTW v2.1	Project	v1.3 Final

3. Test Plan

3.1 Test Environment

Level	Environment	Database	Special Setup
L1 Unit	Developer machine / GitHub Actions; .NET 8, Angular 17 and Flutter 3 test runners	No external DB; dependencies mocked or in-memory as appropriate	Run xUnit/Moq, Jasmine/Karma and flutter_test with static analysis.
L2 Integration (incl. Background Jobs)	Local ASP.NET Core API, Postman/PowerShell HTTP harness and xUnit test runner	Current local development database for API execution; EF Core InMemory for service-integration tests. Test data was created and cleaned through APIs; no direct database edits were used for regression execution.	Disposable Admin, Warehouse, Sales and Auditor sessions; credentials supplied through local environment variables and excluded from evidence. Required role, concurrency and fault-injection scenarios were completed with controlled local fixtures and evidence.

Level	Environment	Database	Special Setup
L3 System / API	Render staging ASP.NET Core 8 API; exact host and build recorded per run	Aiven MySQL 8 staging with SSL required	Warm Render instance; seed normal and 10,000-record performance data; retain Serilog/Hangfire/Aiven evidence.
L3 E2E (Web / Mobile)	Angular Web staging plus controlled Flutter Android APK and physical/simulated ESP32/HX711	Same controlled staging DB	Current Chrome/Edge/Safari; approved Android devices; QR, BLE, FCM, weak-network and reconnect profiles.
L4 UAT	Staging Web/API/APK used by Tuan May or mentor representative	Resettable UAT dataset separated from production	Realistic rice lots, warehouses, debt, sales, returns and physical BLE scale; record build/device versions.

3.2 Test Data

a. Minimum / Target Test Accounts

Username	Role	Password	Account Status	Purpose
qa_admin	Administrator	Stored in local secret/environment ; not displayed	Verified for API test	Authentication, administration and unrestricted verification
qa_owner	Warehouse Owner	Stored in local secret/environment ; not displayed	Verified for selected RBAC tests	Warehouse approval, inventory and stock operations
qa_procurement	Procurement Staff	Not provisioned	Covered by L2 controlled session	Purchase schedule, receipt and farmer payable tests
qa_warehouse	Warehouse Staff	Not provisioned	Covered by L2 controlled session	Putaway, stocktake, packing and transfer tests
qa_milling	Milling Staff	Not provisioned	Covered by L2 controlled session	Milling and yield tests
qa_sales	Sales Staff	Not provisioned	Verified for API test	Sales and receivable debt tests
qa_auditor	Auditor / Internal Reviewer	Not provisioned	Verified for API test	Read-only audit verification; requires explicit role/permission seed

b. Minimum Domain Data

Entity	Count	Required State	Purpose
Warehouses / locations	2 warehouses	Zones and columns include empty, partially occupied, full and priority 1-3 states	Put-away, reservation, stocktake and transfer coverage.
Farmers / purchase documents	3 farmers; multiple schedules/receipts	Pending, confirmed, partially paid and payable-debt states	Procurement, debt, CaptureRequestId / operation key and lot creation coverage.
Paddy/rice lots and inventory	At least 6 lots	Available, reserved, in-process, quarantined and exhausted states across locations	Traceability, concurrency and negative-availability prevention.
Quality inspections	At least 4 records	Pass, warning and quarantine-triggering moisture/quality values	Quality rules, preservation alerts and release controls.
Milling orders	At least 3 orders	Draft, processing and completed with yield/cost inputs and outputs	Yield, costing, duplicate completion and traceability.
Customers / sales / debts	Retail and wholesale customers	Immediate payment, credit, overdue and partial fulfilment states	Sales, reservation, receivable debt and reminders.
Outbound / returns	At least 2 outbound and 2 return records	BLE/manual packing, passed return and quarantined return	Dispatch, reverse logistics and inventory consistency.
Background -job data	Eligible and ineligible records per job	Low stock; intake bottleneck; quality recheck due/not due; long-stored/high-moisture/mould-risk lots; unsent FCM logs; expired/non-expired sessions and verification tokens; payable/receivable debts before, at and after the reminder threshold	JOB-01..JOB-07 eligibility, boundary, failure, deduplication and idempotency tests.

3.3 Test Milestones

Milestone	Target Date	In-charge	Acceptance Criteria
<i>M1 - L1 Unit Complete</i>	Weeks 6, 8, 10; final evidence Week 11	Component developers + Nguyễn Đình Đức	All recorded L1 execution cases complete; Critical/High-priority and state/business-rule tests pass; no open Critical/High L1 defect; CI shows no regression in previously passing L1 cases.

Milestone	Target Date	In-charge	Acceptance Criteria
<i>M2 - L2 Integration Complete</i>	Reports in Weeks 7, 9 and 11	Nguyễn Đình Đức + developers	All executable L2 API, transaction, JOB-01..JOB-07, DataChangeHub and idempotency cases complete with evidence and no stock/debt/ledger inconsistency. Cases requiring unavailable isolated fixtures (including applicable concurrency or infrastructure-failure scenarios) may remain Blocked only when the limitation and closure action are documented and accepted before the applicable L3/release gate.
<i>M3 - L3 System Complete</i>	Week 11 smoke; Week 12 full suite/gate	Nguyễn Đình Đức + Dương Nguyễn Danh Nhân	Critical/High-priority L3 System/API and Web/Mobile business-flow cases pass; no release-blocking workflow remains Failed or Blocked without an approved waiver; applicable NFR/security gates meet baseline; final open defects < 5 and fix rate >= 95%.
<i>M4 - L4 UAT Complete</i>	Week 13	Tuan May / mentor representative	All P1 UAT scripts pass; no critical business blocker; written acceptance decision recorded.
<i>M5 - Closing / Go-Live Readiness</i>	Week 14	Project Lead + StockLite team	Render/API/Web/APK smoke passes; backup, monitoring, warm-up and rollback plan confirmed; milestone records complete.

4. Test Cases

4.1 Test Case Workbooks

Test Level	File	Primary Tool	Contents
<i>L1 - Unit Test</i>	SU26_StockLite_Report_5.1_UnitTests_L1.xlsx	xUnit/Moq; Jasmine/Karma; flutter_test	Backend services, Angular components and Flutter widgets; totals maintained in workbook.
<i>L2 - Integration Test</i>	SU26_StockLite_Report_5.2_IntegrationTests_L2.xlsm	Postman + xUnit + Docker/MySQL	API/layer, screen/input-validation and dedicated background-job sheets.
<i>L3 - System Test</i>	Report 5.3_SystemTests_L3_StockLite.xlsx	Postman + k6/JMeter + browser/device runs	System/API flows and Web/Mobile/Cross-Platform business-flow coverage in one L3 workbook; no separate E2E workbook.
<i>L4 - UAT</i>	SU26_StockLite_Report_5.4_UAT.xlsx	Manual business-language scripts	SC-01..SC-14 acceptance scenarios and sign-off evidence.
<i>Report 5 suite</i>			Grand total maintained from the registered level workbooks.

4.2 Requirements Coverage Matrix

Legend:  indicates planned coverage at L1-L3, not an executed result; actual execution remains Not Run/Pass/Fail in Section 5.1 and the level workbooks. L4 UAT is traced through SC-01..SC-14 business-flow scripts, so FT-level L4 cells are '-' by design.

FT-ID	Feature Name	L1 Unit	L2 Integ.	L3 System	L4 UAT
FT-01	Paddy Procurement Scheduling	☒	☒	☒	-
FT-02	Paddy Purchase Receipt & Payable Debt	☒	☒	☒	-
FT-03	Farmer / Paddy-Seller Master	☒	☒	☒	-
FT-04	Lot-Column-Zone Warehousing	☒	☒	☒	-
FT-05	Quality Management & Preservation	☒	☒	☒	-
FT-06	Quarantine & Sale Lock	☒	☒	☒	-
FT-07	Milling Order (Paddy-to-Rice Conversion)	☒	☒	☒	-
FT-08	Dynamic Milling Yield Configuration	☒	☒	☒	-
FT-09	Lot-Based Product Costing	☒	☒	☒	-
FT-10	Multi-State Inventory	☒	☒	☒	-
FT-11	Lot Traceability	☒	☒	☒	-
FT-12	Column/Lot Outbound Strategy & Put-away	☒	☒	☒	-
FT-13	Stocktake by Zone/Column/Lot	☒	☒	☒	-
FT-14	Rule-Based Quality / Loss / Shrinkage Alerts	☒	☒	☒	-
FT-15	Audit Trail / Activity Log	☒	☒	☒	-
FT-16	Direct Call-In Sales Order (requires milling)	☒	☒	☒	-
FT-17	Scheduled Wholesale / Contract Sales	☒	☒	☒	-
FT-18	Customer & Receivable Debt	☒	☒	☒	-
FT-19	Packing/Weighing (BLE/manual) & Outbound Dispatch	☒	☒	☒	-
FT-20	Returns & Quality Handling (Reverse Logistics)	☒	☒	☒	-
FT-21	Role-Based Access Control	☒	☒	☒	-
FT-22	Multi-Warehouse & Internal Transfer	☒	☒	☒	-
FT-23	Rice Management Reports	☒	☒	☒	-
FT-24	Mobile Field Data-Entry App	☒	☒	☒	-
FT-25	BLE Scale Integration (ESP32/HX711)	☒	☒	☒	-
FT-26	Customer Feedback & Quality Traceability	☒	☒	☒	-

FT-ID	Feature Name	L1 Unit	L2 Integ.	L3 System	L4 UAT
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5. Test Reports

5.1 Summary Table

Round	Metric s	L1 Unit	L2 Integ.	L3 System	L4 UAT	Total
Round #1	Total	1,153	905	Planned	Planned	Managed in level workbooks
	Passed	1,153	843	Not Run	Not Run	Not Run
	Failed	0	62 Fail / 0 Blocked	Not Run	Not Run	0 recorded at L1; remaining levels Not Run
Round #1	Pass Rate	100%	93.1% overall (93.6% Controller/API)	Target 100% of executed release-gate cases	Not Run	L1: 100%; overall pending L2-L4 execution

5.2 Defect Register

a. Test Levels's Defects

Metric	Review	L1 Unit	L2 Integ.	L3 System	L4 UAT	Total
Total Defects	Not Assessed	0 recorded from current L1 execution	48 distinct BUG IDs identified from 62 failed execution cases	Not Assessed	Not Assessed	Not Assessed
Critical Defects Identified	Not Assessed	0	16 Critical BUG IDs identified; central tracker status pending reconciliation	Not Assessed	Not Assessed	14 identified; open/closed status pending reconciliation
High / Medium Defects Identified	Not Assessed	0 recorded from current L1 execution	30 High + 2 Medium BUG IDs identified	Not Run	Not Run	30 identified

Metric	Review	L1 Unit	L2 Integ.	L3 System	L4 UAT	Total
Open High/Critical Defects	Not Assessed	0	Not verified against the central defect tracker; execution status and BUG evidence are available in Report 5.2.	Not Assessed	Not Assessed	Release target: 0 Critical/High open; total final open defects < 5

b. Work Product's Defects

Product	Review	L1 Unit	L2 Integ.	L3 System	L4 UAT	Total
<i>Software Package</i>	Not Assessed	No Critical/High defect recorded from current L1 execution	48 distinct L2 BUG IDs from 62 failed cases; 0 cases remain Blocked.	Not Assessed	Not Assessed	Not Assessed
<i>Report 1 (V&S)</i>	Not Assessed	-	-	-	-	Not Assessed
<i>Report 2 (Plan / Tracking)</i>	Defect tracking inconsistency identified	-	-	-	-	Not Assessed
<i>Reports 3-5 and test workbooks</i>	Documentation consistency issues identified and partially corrected	L1 workbook reviewed and execution results recorded	Report 5.2 reviewed	L3 System/API and Web/Mobile business-flow cases designed in Report 5.3; execution pending	Not Assessed	Not Assessed